

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 6: SHAPE MODELS AND SHAPE RECOGNITION

SLO 1: Describe various shape models used for representing object shapes in digital images.

WORD MATCH SOLUTIONS

Shape Model	Correct Match / Description
Boundary Model	A (Represents the shape using the outer contour)
Region Model	B (Represents the shape using all pixels inside the object)
Skeleton Model	C (Represents the shape using a thin central structure)
Graph Model	D (Represents shape using nodes and connections)

APPLICATION THINKING SOLUTIONS

Application	Suitable Shape Model
Character recognition (OCR)	Boundary Model / Skeleton Model (Captures character stroke outlines and skeletal topologies for invariant recognition)
Medical structure analysis	Region Model (Analyzes internal region pixels, areas, and tissue mass characteristics)
Road network detection	Skeleton Model / Graph Model (Represents thin line-like structures, junctions, and connections between roads)